Table of Basic Integrals

Basic Forms

$$\int x^n dx = \frac{1}{n+1}x^{n+1}, \quad n \neq -1$$
$$\int \frac{1}{x}dx = \ln|x|$$
$$\int udv = uv - \int vdu$$
$$\int \frac{1}{ax+b}dx = \frac{1}{a}\ln|ax+b|$$

Integrals of Rational Functions

$$\int \frac{1}{(x+a)^2}dx = -\frac{1}{x+a}$$
$$\int (x+a)^n dx = \frac{(x+a)^{n+1}}{n+1}, n \neq -1$$
$$\int x(x+a)^n dx = \frac{(x+a)^{n+1}((n+1)x-a)}{(n+1)(n+2)}$$
$$\int \frac{1}{1+x^2}dx = \tan^{-1}x$$
$$\int \frac{1}{a^2+x^2}dx = \frac{1}{a}\tan^{-1}\frac{x}{a}$$
$$\int \frac{x}{a^2+x^2}dx = \frac{1}{2}\ln|a^2+x^2|$$
$$\int \frac{x^2}{a^2+x^2}dx = x - a\tan^{-1}\frac{x}{a}$$
$$\int \frac{x^3}{a^2+x^2}dx = \frac{1}{2}x^2 - \frac{1}{2}a^2\ln|a^2+x^2|$$
$$\int \frac{1}{ax^2+bx+c}dx = \frac{2}{\sqrt{4ac-b^2}}\tan^{-1}\frac{2ax+b}{\sqrt{4ac-b^2}}$$
$$\int \frac{1}{(x+a)(x+b)}dx = \frac{1}{b-a}\ln\frac{a+x}{b+x}, \ a \neq b$$
$$\int \frac{x}{(x+a)^2}dx = \frac{a}{a+x} + \ln|a+x|$$
$$\int \frac{x}{ax^2+bx+c}dx = \frac{1}{2a}\ln|ax^2+bx+c| - \frac{b}{a\sqrt{4ac-b^2}}\tan^{-1}\frac{2ax+b}{\sqrt{4ac-b^2}}$$

Integrals with Roots

$$\int \sqrt{x-a}\,dx = \frac{2}{3}(x-a)^{3/2}$$
$$\int \frac{1}{\sqrt{x\pm a}}dx = 2\sqrt{x\pm a}$$
$$\int \frac{1}{\sqrt{a-x}}dx = -2\sqrt{a-x}$$
$$\int x\sqrt{x-a}\,dx =$$
$$\begin{cases} \frac{2a}{3}(x-a)^{3/2} + \frac{2}{5}(x-a)^{5/2}, & \text{or} \\ \frac{2}{3}x(x-a)^{3/2} - \frac{4}{15}(x-a)^{5/2}, & \text{or} \\ \frac{2}{15}(2a+3x)(x-a)^{3/2} \end{cases}$$
$$\int \sqrt{ax+b}\,dx = \left(\frac{2b}{3a} + \frac{2x}{3}\right)\sqrt{ax+b}$$
$$\int (ax+b)^{3/2}\,dx = \frac{2}{5a}(ax+b)^{5/2}$$
$$\int \frac{x}{\sqrt{x\pm a}}dx = \frac{2}{3}(x \mp 2a)\sqrt{x\pm a}$$
$$\int \sqrt{\frac{x}{a-x}}\,dx =$$
$$-\sqrt{x(a-x)} - a\tan^{-1}\frac{\sqrt{x(a-x)}}{x-a}$$
$$\int \sqrt{\frac{x}{a+x}}\,dx =$$
$$\sqrt{x(a+x)} - a\ln\left[\sqrt{x} + \sqrt{x+a}\right]$$
$$\int x\sqrt{ax+b}\,dx =$$
$$\frac{2}{15a^2}(-2b^2+abx+3a^2x^2)\sqrt{ax+b}$$
$$\int \sqrt{x(ax+b)}\,dx =$$
$$\frac{1}{4a^{3/2}}\left[(2ax+b)\sqrt{ax(ax+b)}\right.$$
$$\left.-b^2\ln\left|a\sqrt{x} + \sqrt{a(ax+b)}\right|\right]$$
$$\int \sqrt{x^3(ax+b)}\,dx =$$
$$\left[\frac{b}{12a} - \frac{b^2}{8a^2x} + \frac{x}{3}\right]\sqrt{x^3(ax+b)} +$$
$$\frac{b^3}{8a^{5/2}}\ln\left|a\sqrt{x} + \sqrt{a(ax+b)}\right|$$
$$\int \sqrt{x^2 \pm a^2}\,dx =$$
$$\frac{1}{2}x\sqrt{x^2\pm a^2} \pm \frac{1}{2}a^2\ln\left|x + \sqrt{x^2\pm a^2}\right|$$
$$\int \sqrt{a^2-x^2}\,dx =$$
$$\frac{1}{2}x\sqrt{a^2-x^2} + \frac{1}{2}a^2\tan^{-1}\frac{x}{\sqrt{a^2-x^2}}$$
$$\int x\sqrt{x^2\pm a^2}\,dx = \frac{1}{3}\left(x^2\pm a^2\right)^{3/2}$$
$$\int \frac{1}{\sqrt{x^2\pm a^2}}dx = \ln\left|x + \sqrt{x^2\pm a^2}\right|$$
$$\int \frac{1}{\sqrt{a^2-x^2}}dx = \sin^{-1}\frac{x}{a}$$
$$\int \frac{x}{\sqrt{x^2\pm a^2}}dx = \sqrt{x^2\pm a^2}$$
$$\int \frac{x}{\sqrt{a^2-x^2}}dx = -\sqrt{a^2-x^2}$$
$$\int \frac{x^2}{\sqrt{x^2\pm a^2}}dx =$$
$$\frac{1}{2}x\sqrt{x^2\pm a^2} \mp \frac{1}{2}a^2\ln\left|x + \sqrt{x^2\pm a^2}\right|$$
$$\int \sqrt{ax^2+bx+c}\,dx =$$
$$\frac{b+2ax}{4a}\sqrt{ax^2+bx+c} +$$
$$\frac{4ac-b^2}{8a^{3/2}}\ln\left|2ax+b+2\sqrt{a(ax^2+bx^+c)}\right|$$
$$\int x\sqrt{ax^2+bx+c}\,dx =$$
$$\frac{1}{48a^{5/2}}\left[2\sqrt{a}\sqrt{ax^2+bx+c}\right.$$
$$(-3b^2+2abx+8a(c+ax^2)) + 3(b^3 -$$
$$\left.4abc)\ln\left|b+2ax+2\sqrt{a}\sqrt{ax^2+bx+c}\right|\right]$$
$$\int \frac{1}{\sqrt{ax^2+bx+c}}dx =$$
$$\frac{1}{\sqrt{a}}\ln\left|2ax+b+2\sqrt{a(ax^2+bx+c)}\right|$$
$$\int \frac{x}{\sqrt{ax^2+bx+c}}dx = \frac{1}{a}\sqrt{ax^2+bx+c} -$$
$$\frac{b}{2a^{3/2}}\ln\left|2ax+b+2\sqrt{a(ax^2+bx+c)}\right|$$
$$\int \frac{dx}{(a^2+x^2)^{3/2}} = \frac{x}{a^2\sqrt{a^2+x^2}}$$

Integrals with Logarithms

$$\int \ln ax\,dx = x\ln ax - x$$
$$\int x\ln x\,dx = \frac{1}{2}x^2\ln x - \frac{x^2}{4}$$
$$\int x^2\ln x\,dx = \frac{1}{3}x^3\ln x - \frac{x^3}{9}$$
$$\int x^n\ln x\,dx =$$
$$x^{n+1}\left(\frac{\ln x}{n+1} - \frac{1}{(n+1)^2}\right), \quad n \neq -1$$
$$\int \frac{\ln ax}{x}dx = \frac{1}{2}(\ln ax)^2$$
$$\int \frac{\ln x}{x^2}dx = -\frac{1}{x} - \frac{\ln x}{x}$$
$$\int \ln(ax+b)\,dx =$$
$$\left(x + \frac{b}{a}\right)\ln(ax+b) - x, a \neq 0$$
$$\int \ln(x^2+a^2)\,dx =$$
$$x\ln(x^2+a^2) + 2a\tan^{-1}\frac{x}{a} - 2x$$
$$\int \ln(x^2-a^2)\,dx =$$
$$x\ln(x^2-a^2) + a\ln\frac{x+a}{x-a} - 2x$$
$$\int \ln\left(ax^2+bx+c\right)\,dx =$$
$$\frac{1}{a}\sqrt{4ac-b^2}\tan^{-1}\frac{2ax+b}{\sqrt{4ac-b^2}} - 2x +$$
$$\left(\frac{b}{2a} + x\right)\ln\left(ax^2+bx+c\right)$$
$$\int x\ln(ax+b)\,dx =$$
$$\frac{bx}{2a} - \frac{1}{4}x^2 + \frac{1}{2}\left(x^2 - \frac{b^2}{a^2}\right)\ln(ax+b)$$
$$\int x\ln\left(a^2-b^2x^2\right)\,dx =$$
$$-\frac{1}{2}x^2 + \frac{1}{2}\left(x^2 - \frac{a^2}{b^2}\right)\ln\left(a^2-b^2x^2\right)$$
$$\int (\ln x)^2\,dx = 2x - 2x\ln x + x(\ln x)^2$$
$$\int (\ln x)^3\,dx =$$
$$-6x + x(\ln x)^3 - 3x(\ln x)^2 + 6x\ln x$$
$$\int x(\ln x)^2\,dx = \frac{x^2}{4} + \frac{1}{2}x^2(\ln x)^2 - \frac{1}{2}x^2\ln x$$
$$\int x^2(\ln x)^2\,dx =$$
$$\frac{2x^3}{27} + \frac{1}{3}x^3(\ln x)^2 - \frac{2}{9}x^3\ln x$$

Integrals with Exponentials

$$\int e^{ax}\,dx = \frac{1}{a}e^{ax}$$
$$\int \sqrt{x}e^{ax}\,dx = \frac{1}{a}\sqrt{x}e^{ax} +$$
$$\frac{i\sqrt{\pi}}{2a^{3/2}}\text{erf}\left(i\sqrt{ax}\right), \text{ where erf}(x) =$$
$$\frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt$$
$$\int xe^x\,dx = (x-1)e^x$$
$$\int xe^{ax}\,dx = \left(\frac{x}{a} - \frac{1}{a^2}\right)e^{ax}$$
$$\int x^2e^x\,dx = \left(x^2-2x+2\right)e^x$$
$$\int x^2e^{ax}\,dx = \left(\frac{x^2}{a} - \frac{2x}{a^2} + \frac{2}{a^3}\right)e^{ax}$$
$$\int x^3e^x\,dx = \left(x^3-3x^2+6x-6\right)e^x$$
$$\int x^ne^{ax}\,dx = \frac{x^ne^{ax}}{a} - \frac{n}{a}\int x^{n-1}e^{ax}\,dx$$
$$\int x^ne^{ax}\,dx = \frac{(-1)^n}{a^{n+1}}\Gamma[1+$$
$$n, -ax], \text{ where } \Gamma(a,x) = \int_x^\infty t^{a-1}e^{-t}\,dt$$
$$\int e^{ax^2}\,dx = -\frac{i\sqrt{\pi}}{2\sqrt{a}}\text{erf}\left(ix\sqrt{a}\right)$$
$$\int e^{-ax^2}\,dx = \frac{\sqrt{\pi}}{2\sqrt{a}}\text{erf}\left(x\sqrt{a}\right)$$
$$\int xe^{-ax^2}\,dx = -\frac{1}{2a}e^{-ax^2}$$
$$\int x^2e^{-ax^2}\,dx =$$
$$\frac{1}{4}\sqrt{\frac{\pi}{a^3}}\text{erf}(x\sqrt{a}) - \frac{x}{2a}e^{-ax^2}$$

Integrals with Trigonometric Functions

$$\int \sin ax\,dx = -\frac{1}{a}\cos ax$$
$$\int \sin^2 ax\,dx = \frac{x}{2} - \frac{\sin 2ax}{4a}$$
$$\int \sin^3 ax\,dx = -\frac{3\cos ax}{4a} + \frac{\cos 3ax}{12a}$$
$$\int \sin^n ax\,dx =$$
$$-\frac{1}{a}\cos ax\ {}_2F_1\left[\frac{1}{2}, \frac{1-n}{2}, \frac{3}{2}, \cos^2 ax\right]$$
$$\int \cos ax\,dx = \frac{1}{a}\sin ax$$
$$\int \cos^2 ax\,dx = \frac{x}{2} + \frac{\sin 2ax}{4a}$$
$$\int \cos^3 axdx = \frac{3\sin ax}{4a} + \frac{\sin 3ax}{12a}$$
$$\int \cos^p axdx = -\frac{1}{a(1+p)}\cos^{1+p} ax \times$$
$${}_2F_1\left[\frac{1+p}{2}, \frac{1}{2}, \frac{3+p}{2}, \cos^2 ax\right]$$
$$\int \cos x\sin x\,dx = \frac{1}{2}\sin^2 x + c_1 =$$
$$-\frac{1}{2}\cos^2 x + c_2 = -\frac{1}{4}\cos 2x + c_3$$
$$\int \cos ax\sin bx\,dx =$$
$$\frac{\cos[(a-b)x]}{2(a-b)} - \frac{\cos[(a+b)x]}{2(a+b)}, a \neq b$$
$$\int \sin^2 ax\cos bx\,dx =$$
$$-\frac{\sin[(2a-b)x]}{4(2a-b)} + \frac{\sin bx}{2b} - \frac{\sin[(2a+b)x]}{4(2a+b)}$$
$$\int \sin^2 x\cos x\,dx = \frac{1}{3}\sin^3 x$$
$$\int \cos^2 ax\sin bx\,dx =$$
$$\frac{\cos[(2a-b)x]}{4(2a-b)} - \frac{\cos bx}{2b} - \frac{\cos[(2a+b)x]}{4(2a+b)}$$
$$\int \cos^2 ax\sin ax\,dx = -\frac{1}{3a}\cos^3 ax$$
$$\int \sin^2 ax\cos^2 bxdx =$$
$$\frac{x}{4} - \frac{\sin 2ax}{8a} - \frac{\sin[2(a-b)x]}{16(a-b)} + \frac{\sin 2bx}{8b} - \frac{\sin[2(a+b)x]}{16(a+b)}$$
$$\int \sin^2 ax\cos^2 ax\,dx = \frac{x}{8} - \frac{\sin 4ax}{32a}$$
$$\int \tan ax\,dx = -\frac{1}{a}\ln\cos ax$$
$$\int \tan^2 ax\,dx = -x + \frac{1}{a}\tan ax$$
$$\int \tan^n ax\,dx =$$
$$\frac{\tan^{n+1} ax}{a(1+n)} \times {}_2F_1\left(\frac{n+1}{2}, 1, \frac{n+3}{2}, -\tan^2 ax\right)$$
$$\int \tan^3 axdx = \frac{1}{a}\ln\cos ax + \frac{1}{2a}\sec^2 ax$$
$$\int \sec x\,dx = \ln|\sec x + \tan x| =$$
$$2\tanh^{-1}\left(\tan\frac{x}{2}\right)$$
$$\int \sec^2 ax\,dx = \frac{1}{a}\tan ax$$
$$\int \sec^3 x\,dx =$$
$$\frac{1}{2}\sec x\tan x + \frac{1}{2}\ln|\sec x + \tan x|$$
$$\int \sec x\tan x\,dx = \sec x$$
$$\int \sec^2 x\tan x\,dx = \frac{1}{2}\sec^2 x$$
$$\int \sec^n x\tan x\,dx = \frac{1}{n}\sec^n x, n \neq 0$$
$$\int \csc x\,dx = \ln\left|\tan\frac{x}{2}\right| =$$
$$\ln|\csc x - \cot x| + C$$
$$\int \csc^2 ax\,dx = -\frac{1}{a}\cot ax$$

$$\int \csc^3 x\,dx =$$
$$-\tfrac{1}{2}\cot x \csc x + \tfrac{1}{2}\ln|\csc x - \cot x|$$
$$\int \csc^n x \cot x\,dx = -\tfrac{1}{n}\csc^n x, n \neq 0$$
$$\int \sec x \csc x\,dx = \ln|\tan x|$$

Products of Trigonometric Functions and Monomials

$$\int x\cos x\,dx = \cos x + x\sin x$$
$$\int x\cos ax\,dx = \tfrac{1}{a^2}\cos ax + \tfrac{x}{a}\sin ax$$
$$\int x^2\cos x\,dx = 2x\cos x + \left(x^2-2\right)\sin x$$
$$\int x^2\cos ax\,dx = \tfrac{2x\cos ax}{a^2} + \tfrac{a^2x^2-2}{a^3}\sin ax$$
$$\int x^n\cos x dx = -\tfrac{1}{2}(i)^{n+1}\left[\Gamma(n+1,-ix)\right.$$
$$\left.+(-1)^n\Gamma(n+1,ix)\right]$$
$$\int x^n\cos ax\,dx =$$
$$\tfrac{1}{2}(ia)^{1-n}\left[(-1)^n\Gamma(n+1,-iax)\right.$$
$$\left.-\Gamma(n+1,ixa)\right]$$
$$\int x\sin x\,dx = -x\cos x + \sin x$$
$$\int x\sin ax\,dx = -\tfrac{x\cos ax}{a} + \tfrac{\sin ax}{a^2}$$
$$\int x^2\sin x\,dx = \left(2-x^2\right)\cos x + 2x\sin x$$
$$\int x^2\sin ax\,dx = \tfrac{2-a^2x^2}{a^3}\cos ax + \tfrac{2x\sin ax}{a^2}$$
$$\int x^n\sin x\,dx = -\tfrac{1}{2}(i)^n\left[\Gamma(n+1,-ix)\right.$$
$$\left.-(-1)^n\Gamma(n+1,-ix)\right]$$
$$\int x\cos^2 x\,dx = \tfrac{x^2}{4} + \tfrac{1}{8}\cos 2x + \tfrac{1}{4}x\sin 2x$$
$$\int x\sin^2 x\,dx = \tfrac{x^2}{4} - \tfrac{1}{8}\cos 2x - \tfrac{1}{4}x\sin 2x$$
$$\int x\tan^2 x\,dx = -\tfrac{x^2}{2} + \ln\cos x + x\tan x$$
$$\int x\sec^2 x\,dx = \ln\cos x + x\tan x$$

Products of Trigonometric Functions and Exponentials

$$\int e^x\sin x\,dx = \tfrac{1}{2}e^x(\sin x - \cos x)$$
$$\int e^{bx}\sin ax\,dx =$$
$$\tfrac{1}{a^2+b^2}e^{bx}(b\sin ax - a\cos ax)$$
$$\int e^x\cos x\,dx = \tfrac{1}{2}e^x(\sin x + \cos x)$$
$$\int e^{bx}\cos ax\,dx =$$
$$\tfrac{1}{a^2+b^2}e^{bx}(a\sin ax + b\cos ax)$$
$$\int xe^x\sin x\,dx =$$
$$\tfrac{1}{2}e^x(\cos x - x\cos x + x\sin x)$$
$$\int xe^x\cos x\,dx =$$
$$\tfrac{1}{2}e^x(x\cos x - \sin x + x\sin x)$$

Integrals of Hyperbolic Functions

$$\int \cosh ax\,dx = \tfrac{1}{a}\sinh ax$$
$$\int e^{ax}\cosh bx\,dx =$$
$$\begin{cases} \dfrac{e^{ax}}{a^2-b^2}[a\cosh bx - b\sinh bx] & a \neq b \\ \dfrac{e^{2ax}}{4a} + \dfrac{x}{2} & a = b \end{cases}$$
$$\int \sinh ax\,dx = \tfrac{1}{a}\cosh ax$$
$$\int e^{ax}\sinh bx\,dx =$$
$$\begin{cases} \dfrac{e^{ax}}{a^2-b^2}[-b\cosh bx + a\sinh bx] & a \neq b \\ \dfrac{e^{2ax}}{4a} - \dfrac{x}{2} & a = b \end{cases}$$
$$\int \tanh ax\,dx = \tfrac{1}{a}\ln\cosh ax$$
$$\int e^{ax}\tanh bx\,dx =$$
$$\begin{cases} \dfrac{e^{(a+2b)x}}{(a+2b)}{}_2F_1\left[1+\dfrac{a}{2b}, 1, 2+\dfrac{a}{2b}, -e^{2bx}\right] & \\ -\dfrac{1}{a}e^{ax}{}_2F_1\left[1, \dfrac{a}{2b}, 1+\dfrac{a}{2b}, -e^{2bx}\right] & a \neq b \\ \dfrac{e^{ax} - 2\tan^{-1}[e^{ax}]}{a} & a = b \end{cases}$$
$$\int \cos ax\cosh bx\,dx =$$
$$\tfrac{1}{a^2+b^2}\left[a\sin ax\cosh bx + b\cos ax\sinh bx\right]$$
$$\int \cos ax\sinh bx\,dx =$$
$$\tfrac{1}{a^2+b^2}\left[b\cos ax\cosh bx + a\sin ax\sinh bx\right]$$
$$\int \sin ax\cosh bx\,dx =$$
$$\tfrac{1}{a^2+b^2}\left[-a\cos ax\cosh bx + b\sin ax\sinh bx\right]$$
$$\int \sin ax\sinh bx\,dx =$$
$$\tfrac{1}{a^2+b^2}\left[b\cosh bx\sin ax - a\cos ax\sinh bx\right]$$
$$\int \sinh ax\cosh axdx =$$
$$\tfrac{1}{4a}\left[-2ax + \sinh 2ax\right]$$
$$\int \sinh ax\cosh bx\,dx =$$
$$\tfrac{1}{b^2-a^2}\left[b\cosh bx\sinh ax - a\cosh ax\sinh bx\right]$$